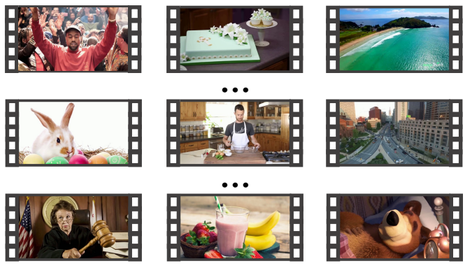
1 Collection & Preparation

Source Video Library

OpenVid-1M / InternVid / OpenVE-3M



Quality Filter

Resolution	FPS	Duration
HD ≥ 720P	FPS ≥ 15	3s ≤ ⏱ ≤ 10s
Clarity	*Smoothness*	*Suitability*

Initial Preparation

Content Categorization



Detailed Recaption



Original Caption: *… man with a beard and sunglasses, wearing a blue shirt and a white baseball cap. He is standing in a grassy area with trees in the background.*

Our Recaption: *… man with a full beard and dark sunglasses, wearing a blue polo shirt and a white mesh baseball cap worn backward. He is positioned in a grassy outdoor area with trees and a paved walkway visible … A black microphone is visible in the lower right corner of the frame, indicating the recording of interview.*

Precise Attributes *Missing Objects* *Spatial Details* *Event Cues*

2 Instruction Generation

Manipulation Tasks



Diversity *Prevalence* *Generality*

Reference Information

Task Initialization

Randomly select one of the six above manipulation tasks.

Source Video

Detailed Recaption

Idea Inspiration

Is the assigned manipulation suitable?

- **Source Object**
- **Target Proposal**
- **Auto Check:** *Is it reasonable and feasible?*
  - 1 *Common Sense*
  - 2 *Event Logic*
- **Open-Vocabulary Manipulation Instruction**

- **Failure Reflection**
- **Task Switch**

3 Video Manipulation

Paradigm A: Video-to-Video (V2V)

Source Input



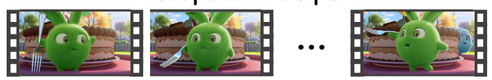
Instruction: *Source Object* *Target Concept* *Temporal Description*

Replace the silver spoon held in the right hand of the green bunny in the video with a silver fork, preserving the original motion trajectory and spatial grounding relative to the bunny's hand and the cakes across the entire timeline.

Video Editing Models (7)

Close-Source Models	Open-Source Models
KLING (Kling-V3-Omni / O1)	VACE (ICCV25)
Wan (Wan 2.2/Video-Edit)	VideoCoF (CVPR26)
runway (Gen-4 Aleph)	VideoPainter (SIGGRAPH25)

Manipulated Output



Paradigm B: Image-to-Video (I2V)

Source Input



Instruction: *Source Object* *Target Concept* *Temporal Description*

Change the material of the wooden dining table in this video into polished granite, maintaining its original shape and spatial grounding relative to the bench seat and window throughout the entire video duration.

First Frame Edit (6)



Canny Video



Canny-Guided I2V Models (3)

Close-Source Models	Open-Source Models
KLING (Kling-V3-Omni)	Wan (Wan 2.2)
	Lightricks (LTX-2)

Manipulated Output



4 Annotation & Verification

Auto Reflection

Source Video | Instruction | Manipulated Video

Manipulation Check

1. *Is the intended manipulation visible?*
2. *Does the result comply with the instruction?*
3. *Are the remaining regions preserved?*

Artifact Grounding

Point Annotation



Source Video



Forged Frame

Target Region *Other Region*

Segment Phrase

Replace the silver toaster with a sleek, stainless steel smart toaster featuring a digital LED display and touch-sensitive control panel, preserving …

"stainless steel smart toaster with LED"

1 *Conciseness* 2 *Commonality*

Forgery Mask (SAM3)



Anomaly Explanation

Forgery Mask | Manipulated Video

Multi-perspective Reasoning



<Select 4 Most Relevant Perspectives>

Human Verification

Task Reasonableness *Manipulation Quality*

Explanation Fidelity *Mask Accuracy*

→ ManiVid